

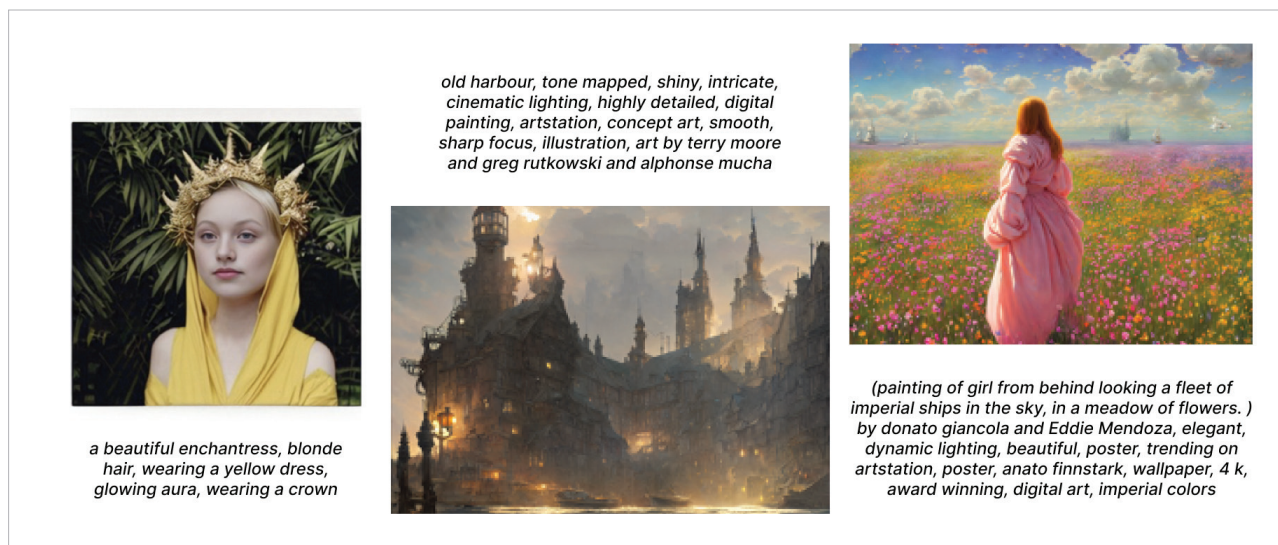


Figure 2: Prompts that produced those images

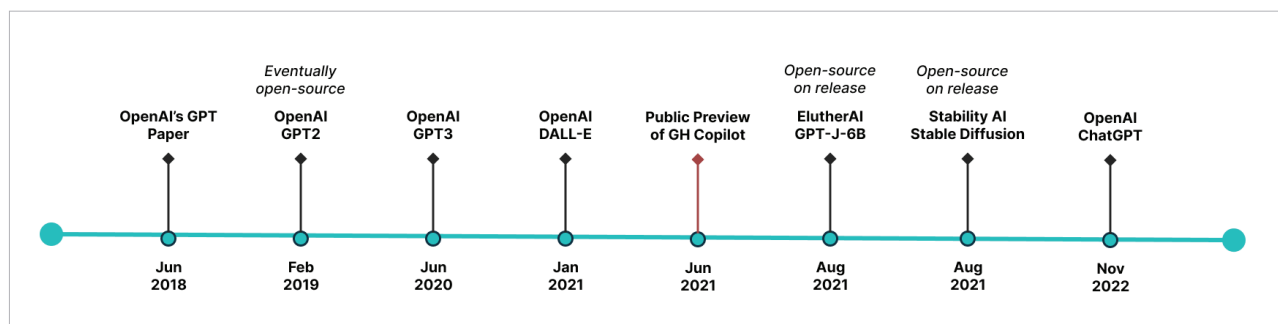


Figure 3: A timeline of events surrounding generative AI

implementations remain closed-source and proprietary.

Programmers were enthused and drawn towards generative AI when GitHub opened up a private preview of GitHub Copilot. Copilot utilised OpenAI Codex — an extension of GPT-3 that is trained on thousands of GitHub repositories. This specialisation allowed it to generate code on-the-fly, infer functionality of code and, more recently, automatically write test cases.

Around this time, EleutherAI created an open source model similar to GPT-3 as well with some impressive demos. It was named 'GPT-J-6B', the J alluding to the Google JAX framework used to develop the model.

However, nothing expanded the open source generative AI scene more

than the open source release of Stability AI's Stable Diffusion. Within hours of its release, technology forums were flooded with inspired individuals expanding the art of the possible. Several developers had built feature-complete, easy-to-use user interfaces and deployment strategies within days of its release. Some even released, modified, and optimised the model to suit specific use cases.

The release of Stable Diffusion was still not the biggest splash that generative AI had created in the world. That's credited to the recent public demo release of ChatGPT, a purpose-built GPT-3 based model capable of holding context-aware conversations and providing an easy-to-use and familiar interface. It reached over one

million users in the short span of a week, a figure that took earlier social networks months and years to achieve.

The keen-eyed among you must have noticed that Stable Diffusion isn't the first open source generative AI model that was launched publicly, yet it somehow broke out of the mould with millions of users. A large chunk of these users self-hosted this model, integrating it deeply into their products and open source projects. So, what did they do differently?

The accessibility and wide adoption can be chalked down to two key factors.

Availability of pretrained weights

Generative AI models are often 'deep neural networks' with millions/billions of parameters, trained over